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ORIGINAL ARTICLE

Gold mining in the Colombian Amazon: empirical insights on the links between sustainability, equity, and power

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Abstract

Gold mining is widespread in the Colombian Amazon and negatively impacts biodiversity, climate, and social equity. While some claim that sustainable gold mining can address these issues, we have analyzed different narratives of sustainable gold mining, focusing on equity and power dimensions. Our research, conducted through participatory observation and interviews across scales between 2022 and 2023, involved local miners, Indigenous representatives, community members, academic experts, NGOs, and government officials at regional, national, and international levels. We found diverging perspectives on sustainable mining leading to conflicts and occasional violence. While the greening of mining involves a corporate focus on distributive equity, traditional and artisanal miners seek support and recognition of their mining rights. Indigenous and other ethnic minorities aim to emancipate themselves from historical epistemic domination and exclusion. We conclude that resolving the conflicts arising from these differences requires questioning the role of corporate mining, and considering diverse values related to the sustainability of gold mining in decision-making processes.

Keywords Colombia · Conflict · Epistemology · Gold mining · Sustainability

Introduction

Climate change threatens the planetary life support systems and humanity's well-being (Rockström et al. 2023; Steffen et al. 2015), and there is a growing urge to transition to more sustainable pathways (IPCC 2023; United Nations Framework Convention on Climate Change 2015). For instance, transitioning to a decarbonized global economy requires scaling up renewable energies like solar and wind power (IPCC 2023), increasing the demand for raw materials, which poses a significant challenge (Lee et al. 2020; Nate et al. 2021). The production of batteries and new technologies for renewable energy requires rare minerals, such as copper, lithium, and nickel, which are mainly found in the Global South (Gielen 2021; Nate et al. 2021; Sovacool et al. 2020). Mining these minerals is highly energy, water, and land-intensive and polluting (Dorn 2022; Hund et al. 2023; Sovacool et al. 2019). This rush for critical minerals may worsen existing inequality, with marginalized groups bearing the burden of the transition (Kalantzakos 2020; Sovacool et al. 2020).

Gold is one of these critical minerals. It is used in hydrogen generators, solar panels, and nanowire-based solar technology (Ismael 2020; Pallas et al. 2020; Sarfraz and Khan 2021). Gold price fluctuations affect investor behavior and capital allocation. Rising gold prices can divert investments away from sectors like clean energy, hindering sustainability transitions. In contrast, stable

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or declining prices can boost investment in green technologies and support economic decarbonization (Bibi et al. 2022). Gold production ranges from artisanal and small-scale mining (ASMG) to large-scale mining dominated by capital-intensive corporations. This leads to significant power asymmetries in the sector, where formal actors disproportionately benefit (Verbrugge and Geenen 2019). ASMG is a vital source of income in rural areas of the Global South (Hilson 2016; Maconachie and Conteh 2021), with around 44.75 million people working in artisanal and small-scale gold mining (ASGM) in 80 countries (Hilson et al. 2021), and producing up to 20% of global supply (Seccatore et al. 2014). However, mining and processing gold require high energy input, emit greenhouse gases, drive deforestation, and cause mercury pollution—primarily in informal operations—and cyanide use in formal mining, both contributing to a negative global footprint (González-González et al. 2021; Igogo et al. 2021; UN Environmental Programme 2018).

To transition toward more sustainable and equitable forms of gold mining, a comprehensive understanding of the social and ecological impact of mining is essential. However, the concepts of sustainability and equity are complex and contested (Leach 2015), because their interpretation varies depending on the context and values of the different actors involved (Caniglia et al. 2020; Leach et al. 2018). Power dynamics play a significant role in determining which interpretations of sustainability prevail (Quimby and Levine 2018) and, thus, how the ecological and social impacts of gold mining are assessed and managed. Understanding the influence of power on sustainability transitions (Boonstra 2016) is crucial for addressing the unequal distribution of benefits and burdens associated with gold mining (Bedford et al. 2020).

Our contribution to the debate on sustainability transitions is to explore how power relations between stakeholders influence their visions of equitable and sustainable gold mining. We focus on the case of gold mining in the Colombian Amazon region, which involves a wide range of actors and has garnered global concern due to its adverse environmental and social effects (UNODC 2021; Vélez-Torres et al. 2018). We conducted a discourse analysis of stakeholders' imaginaries on sustainable and equitable gold mining, focusing on regional, national, and international actors in Colombia and Europe, such as Indigenous people, researchers, companies, government agencies, and NGOs.

The paper is organized as follows: first, we introduce our conceptualization of power and equity, followed by a general overview of mining policies and sustainability in Colombia. Next, we describe our research methods, including the data collection and analysis. The results section presents the narratives of different actors on sustainable and equitable gold mining, which we analyze using equity and power theories. Finally, we discuss their implications on equitable and sustainable transitions and present our conclusions.

Theoretical framework

The study of power is fundamental for understanding human behavior, assigning responsibility, and evaluating institutional performance in terms of justice (Haywarda and Lukesb 2008; Sayer 2012). However, due to its polymorphic nature (Elias 2012), scholars tend to debate different dimensions of power (Haugaard 2020; Hayward and Lukes 2008; Rowbotham 1998; Sayer 2012). In the next section, we outline how we use Haugaard's (2020) typology of power to understand conflicts and power, and then explain the equity aspects covered in our study. Haugaard distinguishes between 1D, 2D, and 3D conflicts, illustrating how power operates via direct domination, structural contestation, and knowledge hierarchies. We apply the tridimensional equity framework—distributive, procedural, and recognitional—to assess justice in gold mining areas. Table 1 summarizes the main theoretical concepts utilized in this paper.

Power-related conflicts

A new vital contribution to the power debate is Haugaard's (2020) typology of power, synthesizing power theory into a cohesive framework (Allen et al. 2021). Haugaard's typology effectively unpacks the complex dynamics in the Colombian Amazon, where various actor groups wield power not just through coercion or institutional authority but also by controlling how meaning is constructed, who is recognized, and who is excluded from visibility. While frameworks like Avelino (2021), Clement (2013), and Patterson et al. (2017) provide useful insights into various forms of power in sustainability transitions and governance, Haugaard's model offers a more relational perspective—highlighting how power manifests through social reproduction, epistemic recognition, and the structuring of everyday interactions.

Haugaard (2020) introduces four dimensions of power (1D, 2D, 3D, and 4D), offering an innovative approach that captures both relational and structural dynamics, making it particularly well suited for analyzing complex issues of social formation. In our study, we use the first three power dimensions to analyze sustainable and equitable practices in gold mining in the Colombian Amazon, excluding the fourth dimension as we did not identify conflicts linked to slavery or social death (see Haugaard 2020). This section summarizes Haugaard's typology with examples from the context of gold mining in Colombia.

Table 1 Summary of the theoretical framework on power conflicts and environmental equity

Theme	Concept	Key definitions/ideas	Example
Power conflicts (Haugaard 2020)	1D conflict	Power-over through coercion; zero-sum relationships	Armed groups forcing miners to pay "protection" fees under threat of violence
	Positive-sum power (1D)	Power relationships where compliance empowers the less powerful actor	Informal mining permits where landowners grant miners access to deposits
	2D structural conflict	Contestation of social structures; challenge to legitimacy	Nasa Indigenous people's civil disobedience reclaiming Cerro Berlin
	2D structural bias	Reinforcement of social structures through habitual behavior	Use of dominant language forms vs. Francia Marquez promoting inclusive, community-rooted language
	2D epistemic conflict and exclusion	Mutual incomprehension or invalidation between different knowledge systems	Indigenous knowledge being dismissed or marginalized in favor of Western views
	3D epistemic domination	Structural privileging of one knowledge system over another	Western biomedical knowledge overshadowing Indigenous healing practices
Tri-dimensional equity	Distributive Equity	Perception, allocation, and sharing of mining burdens and benefits	How benefits (e.g., profits) and harms (e.g., environmental damage) are distributed among mining stakeholders
	Procedural equity	Inclusiveness, transparency, and conflict resolution in decision-making	Who participates in decisions about mining, and how conflicts over resource access are resolved
	Recognitional equity	Respect for cultural and epistemological differences; rights to self-determination	Valuing Indigenous and local knowledge and rights in gold mining decision-making

This table summarizes the main concepts drawn from Haugaard's (2020) typology of power and recent tri-dimensional equity scholarship used in this study

Haugaard (2020) defines *1D conflicts* as disagreements that are produced when more powerful actors use coercion to make less powerful actors do things they otherwise would not do. This power relationship is known as a *zero-sum power*, with the gains of the dominant actors entirely at the expense of the less powerful. For instance, in Colombia, gold miners pay armed groups for "protection." Although these armed groups consider this payment as an exchange for security, if miners do not pay, they face violence and sometimes death. *1D structured conflicts* are *1D conflicts* that occur around social structures, such as the democratic process between mining corporations and trade unions.

Positive-sum power relationships can exist in *1D* power where the more powerful actor has power over the less powerful one. Compliance by the less powerful actor can lead to power-to (act) of the less powerful actor, thus empowering them. Examples are informal mining permits where landowners grant miners access to gold deposits in exchange for a share of the proceeds. Landowners hold power by controlling access to the deposits and mining revenues, while miners benefit by selling the remaining gold in the local market. This legitimate form of power is described as a *positive-sum power relationship* in which both parties gain something, despite the more powerful usually gaining more.

Two-dimensional *(2D) structural conflicts* arise when social structures are contested through resistance or alternative practices. *2D structural bias* refers to how social structures are reinforced when individuals reproduce expected behaviors, embedding specific advantages or disadvantages into the social order (Haugaard 2020). While bias stabilizes these structures, conflicts emerge when individuals try to challenge them. A 2D *structural conflict* is thus created when social structures are contested. *2D structural conflicts* imply a fundamental disagreement over the social structure or rules of the game, with actors avoiding reproducing the given social order.

Compared to *1D structured conflicts*, in *2D structural conflicts,* the authority of powerful actors breaks down and is perceived as illegitimate. The breakdown of authority can lead to dominant actors using coercion or violence to exercise power-over. For instance, on July 17, 2012, the Nasa Indigenous people took control of Cerro Berlin, a strategic military point above Toribio, in Cauca, expelling the Colombian military. The Nasa sought to reclaim their territory and remove armed actors, whose presence threatened their autonomy. This action sparked a national debate, with some supporting Indigenous autonomy and others defending

the Colombian state's sovereignty. Former president Juan Manuel Santos condemned the actions as criminal, threatening to use the state's legal power to restore order (Peluffo and Gilma 2016). The Nasa indigenous people's civil disobedience is an example of a *2D structural conflict* that challenges the Colombian state's authority over the Cauca department by declaring it an autonomous territory where all forms of state and non-state-related violence were rejected.

According to Haugaard (2020), what counts as reasonable in society determines the meaning and scope of authority. This can be measured by actors' responses to others' exercise of power. When a social order is seen as natural, it becomes unquestionable and stable, with people reproducing it without considering its construction or impact on society (*2D structural bias*). Conversely, societal change occurs when a smaller group agrees on an initially unreasonable idea, creating a common understanding. When a larger group adopts this idea, broader societal change happens. For example, Francia Marquez, the first Afro-descendant woman to become vice president in Colombia, is known for using her local community's terminology in public. She uses neutral nouns for feminine and masculine forms (e.g., *mayores y mayoras, los y las nadies*) and aims to make her community's private language public using inclusive language, challenging the "reasonable" way of speaking nationally.

When social actors rely on different knowledge systems, their statements may appear incomprehensible or invalid to each other, leading to epistemic exclusion *(2D exclusion)* or direct epistemic conflict *(2D epistemic conflict)*. In contrast, *3D epistemic domination* occurs when one knowledge system is structurally privileged over another, shaping what is valid knowledge (Haugaard 2020). In Colombia, Western knowledge systems have historically dominated Indigenous epistemologies, a dynamic that continues in contemporary institutions. For instance, in Putumayo, Indigenous medical practices like the use of yagé (ayahuasca) are often dismissed as superstition and excluded from formal public health policies (Campo 2021). This reflects how colonial hierarchies of knowledge persist, sidelining locally grounded systems of care in favor of Western biomedical models.

Tri-dimensional equity in environmental justice

Studies of power are used to evaluate both people and institutions in terms of justice. Environmental justice is used to perform such evaluations, focusing on the unequal distribution of environmental goods and burdens (Murdock 2020; Taylor 2000). The term emerged in the 1980s in the US, associated with communities of color, mainly African-Americans, protesting commercial waste disposal (Murdock 2020). Since then, environmental justice has become a crucial ground for activism and resistance worldwide (Carmin and Agyeman 2011; Sikor and Newell 2014). More recent theoretical developments within the field have conceptualized environmental justice in distributive, procedural, and recognitional equity, three mutually interacting and reinforcing dimensions (Pascual et al. 2014; Sikor and Newell 2014). Although the terms "equity" and "justice" are often used interchangeably when referring to these dimensions, we use "equity" to highlight the connection between environmental equity and sustainability (Hamann et al. 2018; Pascual et al. 2014; Zafra-Calvo et al. 2020). Environmental equity is understood as the result of environmental justice efforts to attain sustainability.

In this research, we use the tri-dimensional conceptualization of equity to study environmental equity in the context of gold mining as follows. We understand *distributive equity* as how burdens and benefits of gold mining are perceived, allocated, and shared among actors in mining territories (Leach et al. 2018; Schlosberg 2013). *Procedural equity* refers to how decisions about mining are made, who should or should not receive the benefits and burdens of gold mining, how inclusive and transparent decision-making is (Leach et al. 2018), to what extent local actors can solve conflicts about perceived and experienced inequity, and to whom they raise concerns for conflict resolution around, for instance, access to mining deposits (Sikor and Newell 2014). *Recognition* refers to the respect of cultural and epistemologic differences, being primarily expressed in the claims of Indigenous and local people to self-determination and respect. Recognition deals with how discursive power suppresses epistemological and cultural diversity (Martin et al. 2016). We define recognition as related to who has a legitimate right to make decisions about resource access and use in mining territories, meaning that the valuation attributed or denied to cultural diversity, knowledge systems, and statutory and customary rights in gold mining-related decision-making (Leach et al. 2018; Martin et al. 2016).

Background

Colombia has sought to protect the Amazon while recognizing the importance of mining as an economic activity of public interest since the 1950s. Gold mining drives forest degradation and mercury pollution in rivers in Colombia, mainly due to informal and illegal mining despite the 2018 national ban of such activities (Diaz et al. 2020; González-González et al. 2021; Krause et al. 2022). Estimates suggest that extracting 1 ton of gold leads to the deforestation of 39.2 hectares (González-González et al. 2021). Approximately 193 tons of mercury are used annually for mining activities in the country (Diaz et al. 2020). ASGM accounts for a large share of Colombia's gold production but is characterized by high rates of informality and illegality, exacerbating

environmental degradation and intensifying social conflicts (Diaz et al. 2020; Rettberg et al. 2020; Specht and Ros-Tonen 2017). The expansion of mining has particularly impacted Indigenous territories and Afro-descendant communities, deepening historical patterns of dispossession and marginalization (Arbeláez-Ruiz 2021; Valdelamar-Villegas and Olivero-Verbel 2020).

Conflicts surrounding gold mining in the Colombian Amazon involve disputes over who can mine, how mining should be conducted, and where mining operations should be located. The Colombian state's limited presence in the Amazon, combined with the power vacuum created by the Revolutionary Armed Forces of Colombia (FARC) demobilization—the de facto state guerrilla group—has allowed informal and illegal mining to flourish, often linked to conflict and territorial disputes. Consequently, various actors, including armed groups, insurgents, drug traffickers, corporations, and informal miners, compete for gold resources (Diaz et al. 2020; Rettberg et al. 2020). Armed groups often control such activities, including dissident FARC factions, criminal networks, and paramilitary successors, which expand the mining frontier and worsen social and environmental impacts. This connection undermines environmental regulations and perpetuates violence and displacement in affected regions (Krause et al. 2022; Massé and Le Billon 2018; Rettberg et al. 2020). In this context, sustainability becomes a contested concept, used by different actors to justify competing claims to territory, legality, and ecological responsibility.

Meanwhile, the national government has broadly criminalized informal and illegal mining, deploying military operations to dismantle illicit gold production (UNODC 2021). However, expanding corporate, formal mining projects have also led to social tensions, as traditional miners face criminalization, struggle to comply with new regulations, and are increasingly displaced from their lands (Le Billon et al. 2020).

Legal and policy frameworks in Colombia reflect these contradictions. Since 1959, environmental conservation policies have been introduced to protect the Amazon (e.g., Constitution 1991; Decree 2811 1974; Law 2 1959). More recently, the Amazon was recognized as a legal subject of rights (Sentence 4360 2018). In parallel, mining has been promoted as a driver of sustainable development under the 1991 Constitution and the 2001 Mining Code (Law 685 2001). This dual legal commitment—to both environmental conservation and mineral extraction—poses inherent contradictions that are challenging to resolve, especially in remote Amazonian areas with limited state oversight (Procuraduría General de la Nación 2024; UNODC 2021).

Against this backdrop, this study examines how different actors—miners, corporations, state agencies, NGOs, and Indigenous organizations—navigate the contested terrain of gold mining in the Colombian Amazon. It analyzes how struggles over resource control, sustainability, and equity unfold in a context of legal pluralism, historical exclusion, and ongoing socio-ecological conflicts.

Methods

Data collection

To investigate the connections between sustainable gold mining, equity, and power, we conducted a 3-month (February–May 2022) and a 6-month (November 2022–May 2023)

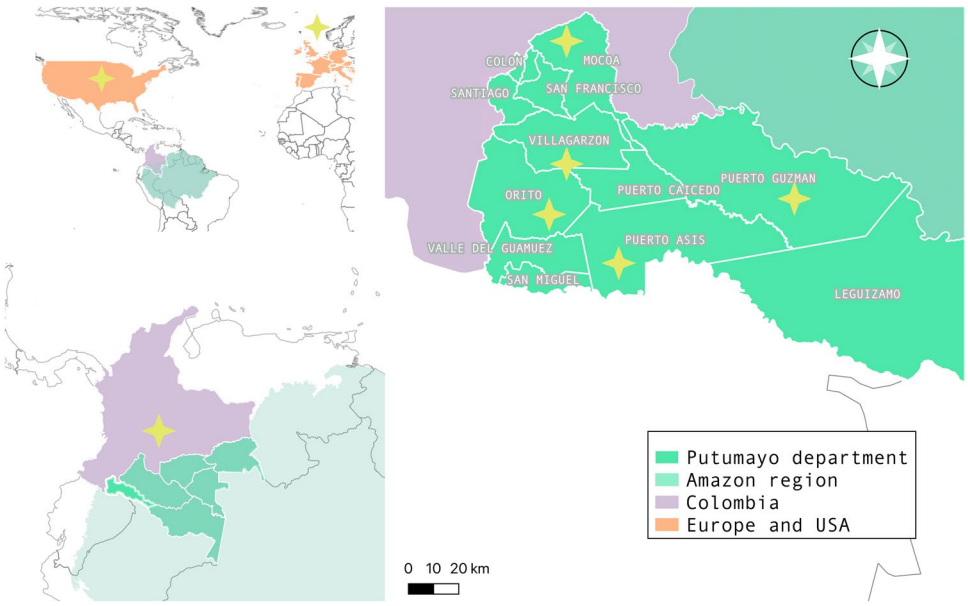


Fig. 1 Locations for data collection. The stars indicate where interviews or participatory observations were conducted

fieldwork trip. We combined participant observation with 20 semi-structured interviews conducted with stakeholders involved in gold mining at the local/regional ($n = 10$), national ($n = 7$), and international ($n = 3$) levels (Fig. 1; Online Resource 1. Table S1).

Initially, we interviewed small-scale miners, social leaders, grassroots organizations, and local government officials. We then expanded to include national state agencies, researchers, NGOs, multilateral organizations, jewelers, refiners, mining companies, ecosystem conservation groups, and central banks until saturation was reached using a snowballing approach (Emerson 2015; Schreier 2018). Interviews were conducted online, using Zoom, for actors based in Europe and USA ($n = 7$) and in-person ($n = 13$) for actors based in Bogotá and the Putumayo department. Interviewees who were Colombians and conducted research at the local or national level, but were based in Europe, were considered to be at the national level. Interviews lasted up to 1 h and were conducted in Spanish ($n = 19$) and English ($n = 1$) (see Online Resource 1. Appendix 1). In accordance with ethical and data protection standards, the names of private companies interviewed have been anonymized. References to corporate actors are therefore made in general terms.

Interviewees were first asked about their understanding of sustainability and its relationship with mining and tri-dimensional equity. We included questions about (1) the positive and negative effects of gold mining (*distributive equity*), (2) the conflicts and citizen participation *(procedural equity*), and (3) cultural diversity in gold mining territories (*recognition*). Furthermore, we sought to understand how interviewees connected three equity dimensions to sustainability and inquired about regional and national strategies for making gold mining more sustainable, the future of gold mining in Colombia, and whether mining should be part of a sustainable future. A complete description of compliance with ethics standards, along with anonymized interviews, is available at https://doi.org/10.6084/m9.figshare.27248430.

Data analysis

We conducted a qualitative content analysis using NVivo 12, following a selective coding approach (Kuckartz 2014). Initial codes were developed deductively based on the research questions and literature on sustainable gold mining, climate change, equity, and power-related conflicts. As analysis progressed, we refined codes inductively to capture emerging themes, categorizing the data into four groups (Business as usual; Geening gold mining; Scaling-up ASGM, and Emancipatory practices in gold mining territories) (see Online Resource 1. Appendix S2). The lead author performed the primary coding and analysis, with co-authors providing feedback at key points to support category development and enhance coherence. We discussed and refined categories collaboratively to strengthen the rigor of the analysis.

To mitigate bias in stakeholder selection and narrative construction, we triangulated interview data with findings from prior document analysis and data collection (Sánchez-García and Wong 2024). We maintained a reflexive approach throughout the research, critically examining our own positionality and the power relationships that influenced our interactions with participants. From this analysis, we derived three ideal–typical narratives—Greening of gold mining, Assisting ASGM, and Emancipatory practices and resistance—which reflect different imaginaries of sustainability, equity, and conflict in gold mining territories. These narratives are heuristic constructs (Stapley et al. 2022); imaginaries are overlapping and dynamic, rather than neatly separated. While interviewees occasionally mentioned the effects of armed conflict and violence, our primary focus is on sustainability imaginaries, based on the belief that sustainability cannot be truly realized amid persistent violent conflict.

Results

In what follows, we present three distinct narratives on sustainable gold mining, each aligned with different sets of actors and imaginaries. The "Greening of gold mining" narrative primarily reflects the perspectives of corporate and state actors, while civil society organizations and grassroots movements are more prominently represented in the "Assisting ASGM" and "Emancipatory practices and resistance" narratives—see Table 2 for a summary. While Table 2 summarizes the main narratives and actor imaginaries, interviewees often described corporate actors and state institutions—such as the Ministry of Energy and Mines, and the Regional Environmental Agency (CAR)—as collaborating to promote extractive interests. The table notes these distinctions, but our analysis highlights how these actors are seen as overlapping in practice rather than being separate.

Greening of gold mining

In the first narrative on sustainable and equitable mining, sustainability is perceived as an equitable intergenerational form of development and implies 'greening,' i.e., reducing the negative environmental impacts of mining. Interviewees—mining companies, refiners, central banks, and the National Police, the National Mining Agency, the Ministry of Environment, the Departmental Government of Putumayo—argue that it is unreasonable to think of gold mining as a sustainable activity, since it implies the depletion of gold over time, and causes the degradation of natural

Table 2 Summary of actors, imaginaries of sustainable mining, proposed actions, and dimensions of equity to be attained in mining territories in the Colombian Amazon for each narrative (greening of gold mining; assisting ASGM; and emancipatory practices and resistance)

Narrative	Actors	Imaginaries of sustainable mining	Proposed actions toward an equitable and sustainable transition	Dimensions of equity to be attained	Challenges and (or) critics
Greening of gold mining	Corporate mining companies, refiners, central banks, national and regional state agencies, such as the National Mining Agency, the Ministry of Environment, and the departmental government of Putumayo	An equitable intergenerational form of development that implies the reduction of the negative environmental impacts of gold mining	(1) Reduce the environmental impacts of gold mining by implementing high-tech and science-based solutions (2) Avoid distributional inequity in mining territories by sharing the economic benefits of gold mining with less powerful actors like artisanal miners	Distributive	(1) Dependency on corporate-driven scientific solutions leads to (3D) epistemic domination and (2D) exclusion of local and traditional mining practices (2) Installed capacities and financing for implementing science-based and high-tech solutions are notably absent in the region
Assisting ASGM	Artisanal, small- and medium-scale miners; national and international NGOs and researchers; and more recently some government agencies at the regional and national level, such as Ministry of Mines and Energy	The ecological conservation and social development can be realized through ASGM	(1) Recognize miners' customary and statutory rights over the land and mining titles (2) Gold miners demand technical and financial assistance and training to comply with legal regulations of the activity to revert (2D) epistemic exclusion (3) Transparency of decision-making in mining territories	Procedural and recognition	(1) Legal mechanisms for assisting artisanal, small- and medium-scale miners do not exist or have not yet been implemented (2) Conflicts between corporate actors and artisanal miners over mining titles (3) Miners' distrust of state institutions
Emancipatory practices and resistance	Indigenous and ethnic minorities, peasant and environmental grassroots organizations, NGOs, municipal governments, such as Mocoa's local government	Incommensurable—good living is underpinned by autonomy	(1) Resistance and emancipation practices to counteract Western (3D) epistemic domination	Recognition	(1) Coercion and repression of alternative viewpoints to development (e.g., Buen Vivir) (2) Epistemic and physical violence against Indigenous and local people (3) Lack of legitimacy of state agencies

processes and ecosystems. Nonetheless, interviewees view mining as essential to human development and modernization in Colombia, believing that modernization cannot be achieved without it (2D structural bias).

One multilateral organization officer, for example, explained that without mining, Colombians would have to forgo modern conveniences like clothing, cars, and infrastructure, reverting to subsistence lifestyles similar to those in isolated Amazonian communities, *"If* [mining] *is not done, we will have to stop wearing clothes, we will* [need] *to stop using cars* [to live] *in the middle of the river* [and] *go hunting or fishing"* (220312_002_Multilateral_organization_officer).

Solutions to overcome the tension between environmental sustainability and the need for human development are based on modes of gold mining that reduce environmental harm and avoid the most common types of social harms and distributive inequities in mining territories. As one corporate refiner summarized, the aim is to ensure that gold supply chains contribute to local development while minimizing long-term environmental impacts and providing fair compensation to miners based on international gold prices: *"What we* [as a company] *want is to* […] *contribute to development and the economy of the miners on the ground and* […] *not have a negative environmental impact* […]" (221003_001_Refiner).

Environmental harm is reduced in this case by developing high-tech solutions based on advanced scientific knowledge and expertise linked to reforestation, safe chemical transportation, and hazardous waste management to avoid pollution. Social harms and distributive inequities can be reduced by: (1) better integrating ASGM practices into formal national and global profit-driven markets or absorbing miners as workers into industrial mining operations; and, (2) minimizing social costs of environmental transformations such as land-use change and mercury pollution by ensuring that women, children, Indigenous peoples, and local communities are not disproportionately affected by mining activities.

These strategies aim to resolve 1D distributive conflicts by improving the material conditions of less powerful actors through their inclusion in formal economic systems. However, respondents acknowledge that greening initiatives primarily serve corporate interests, allowing national and transnational mining companies to secure profits while maintaining the status quo. Implementing low-cost mitigation and compensation enables companies to secure capital accumulation while projecting an image of environmental responsibility. Moreover, skilled workers at the national and, to some extent, regional level—primarily based in Bogotá and other critical urban centers like Mocoa—benefit directly and indirectly from well-established mining industries, both through employment in technical roles and by securing positions in state mining and environmental agencies (e.g., National Mining Agency, Ministry of the Environment, Ministry of the Interior, and CARs) at the national and regional scale.

However, the greening narrative reproduces a 2D structural bias by positioning science-based, large-scale industrial mining as the only viable pathway to sustainable development. ASGM is depicted as costly, unscientific, challenging to implement and monitor, and therefore inefficient. This results in (3D epistemic) domination, where Western technical knowledge systems are privileged over Indigenous, local, and traditional knowledge regarding gold extraction and environmental stewardship.

Assisting ASGM

Interviewees argue that through the assistance of traditional and small-scale mining, social sustainability can be attained better as it allows for less polluting and resource-consuming forms of economic growth and local development. Respondents who voice this narrative find that, although mining expansion entails inevitable trade-offs between ecosystem conservation and the exploitation of gold, social development can be attained through ASGM practices. As a result, local and artisanal miners challenge corporate mining by appealing to their customary and statutory rights over land and resources in mining territories to defend their subsistence economies based on gold extraction.

A government official explained that many small-scale miners, once unaware of the formal mining title process, now mobilize constitutional legal instruments to secure rights over their traditional territories as they perceive growing threats from external actors: *"*[…] *he feels that something is being taken away from him, then he turns to the different* [legal] *instruments that we have in the Constitution* [to attain the mining title]*"* (220426_001_Goverment_officer).

Knowing the importance of controlling gold reservoirs to reproduce their subsistence economy, artisanal and small-scale miners use national legal instruments, such as mining title formalization procedures established in the Colombian Mining Code, which stipulates in Article 16 the principle of "first in time, first in law" to prioritize mining rights. Miners know that, despite their exclusion from decision-making arenas (2D exclusion), they will be formally granted mining rights if they plead first.

Yet, acquiring mining titles and complying with environmental regulations is highly challenging for small-scale miners, as the formalization processes are designed around corporate models. A researcher noted that while the government pressures miners to formalize, the process reflects corporate paradigms, disregarding traditional extraction scales and cultural values*:"*[T]*his formalization the community has taken it very badly because it is a formalization that comes in the corporate terms and the community wants*

a formalization that understands the traditional paradigm" (220317_001_Researcher).

Although formalization can shield miners from prosecution, interviewees argue that it is insufficient to assist artisanal and small-scale miners as it does little to change imaginaries of greener mining under the corporate vision and address miners' (2D) exclusion. On the contrary, this notion reinforces assumptions that miners are not knowledgeable voices capable of contributing to decision-making. High-tech and science-based regulations for ASGM often reinforce the epistemic exclusion (2D exclusion) of traditional and non-corporate forms of mining.

While the Ministry of Energy and Mines has made some efforts, which resulted in, for example, the Law 1658 of 2013 and Law 2250 of 2022, to recognize traditional mining and formalize ASGM (e.g., the Plan Único de Legalización y Formalización Minera), interviewees stress that top–down initiatives are poorly implemented, leaving local miners without essential technical and financial support. Besides, the lack of transparency and inclusiveness in the adjudication of mining titles has deepened and widened conflicts between local inhabitants, miners, mining companies, and state agencies. Local miners reported that companies often acquire mining titles through fraudulent bidding processes, despite prior claims filed by artisanal miners (220316_001_Miner_and_social_leader). This lack of transparency leads to protracted (1D structured) conflicts as small-scale miners resist displacement and refuse to abandon their activities even when facing corporate land claims.

Miners increasingly see transnational corporations as illegitimate, because they lack a social license to operate in local arenas. Lack of transparency in the titling process causes enduring conflicts between local miners and mining companies, eroding trust in state institutions, notably the national government and the National Mining Agency. Frustrated by perceived injustices, some artisanal miners have legally challenged the National Mining Agency to revoke corporate titles and acknowledge their customary rights.

In other regions like Mitú, in Vaupés, corporations have tried to settle disputes with miners by offering higher wages (220316_001_Miner_and_social_leader). However, many miners reject wage work, seeing it as a loss of autonomy and increased corporate dependency. Interviewees suggest that this strategy fails to legitimize mining companies locally and reinforces a zero-sum power dynamic between large-scale and small-scale mining actors.

Emancipatory practices and resistance: challenging narratives of development

The representatives of a range of local and regional Indigenous, ethnic, peasant, and environmental grassroots organizations, NGOs, and municipal governments we interviewed are critical of simultaneously mining gold and protecting the ecological integrity of the Amazon rainforests. Respondents argue that the gold industry profits and other economic benefits will never compensate for the environmental and cultural loss, and, as a result, all forms of for-profit mining should be banned in the Amazon. Indigenous people and environmental organizations consider mining a threat to all life forms in the Amazon region. These groups challenge Western imaginaries of modernization as a form of good living, expressing a decolonial perspective that prioritizes alternative visions of sustainability rooted in their own territorial and cultural framework.

One Indigenous leader explained that companies often promise improved livelihoods—better houses, cars, and incomes—without disclosing the environmental destruction that mining would cause. She reflected on how, out of necessity and lack of information, many communities allowed mining projects, only to realize too late that they had *"exchanged water sources for cement houses."* Later, she described the emotional impact of seeing mining concessions mapped across their territories, expressing profound sadness and a sense of dispossession (220317_002_Indigenous_leader).

In this narrative, sustainability entails the emancipation of Indigenous, ethnic, peasant, and other local people's ways of living by guaranteeing local autonomy to hold the power-over their territories. Indigenous, ethnic, peasant, and environmental organizations claim that Buen Vivir represents the regional perception of sustainability. Buen Vivir in South America prioritizes the common good over individual well-being, emphasizing respect for all living beings and the planet as Mother Earth (Estermann 2012; Villalba 2013). It opposes Western modernization's focus on growth and development fueled by commodity accumulation (Huanacuni 2010). In this sense, Buen Vivir constitutes a decolonial framework that challenges dominant development paradigms and redefines sustainability through Indigenous epistemologies.

The (3D) epistemic domination of the Western ontology over other knowledge systems has resulted in the exclusion of Indigenous and ethnic minorities, as explained by a European NGO officer: *"I believe that there is also a certain type of racism* [and] *classism, that you as an uptight person from Bogota* […] *know what the Afro or Indigenous people need."* (221011_001_International_NGO_officer). During the interview, the speaker emphasizes that people in Bogotá have an underlying epistemic superiority in taking action-oriented solutions in local territories.

Indigenous, Afro-descendant, and other grassroots organizations use their collective power to counteract hegemonic imaginaries of sustainability linked to the greening of gold mining, demanding the fulfillment of the constitutional

mandate to protect ethnic and cultural diversity in the country (Constitution 1991, Art. 7–8). However, the idea of modernity and development as the only way of living results in a (2D) epistemic conflict between emancipatory practices and development imaginaries, with powerful actors, such as mining companies and government officials, operating in the national and regional arenas.

The (2D) structural conflicts linked to (3D) epistemic domination and a lack of legitimacy of state agencies result in episodic violence and confrontations in mining territories. Yet, powerful actors continue to coerce and repress counter-imaginaries of development in the region. In response, Indigenous and local organizations increasingly articulate a decolonial resistance that combines territorial defense with the assertion of epistemic autonomy.

Indigenous people and other local inhabitants are aware of how domination results in epistemic and life-threatening physical violence. Nowadays, they prefer to resist through national and international legal mechanisms (e.g., Free, Prior, and Informed Consent), aiming to down-scale (2D) structural conflicts to (1D) structured legal conflicts. As one Indigenous leader described, rather than directly confronting companies, communities now resist through "legal tools" that have reduced the frequency of assassinations and violent retaliation: *"*[…] *with* [legal] *tools* […] *that is when the killing of people stops a little,* [which means] *the leaders no longer confront* [the companies] *directly to demand respect"* (220317_002_Indigenous_leader).

Discussion

Rising global gold demand will likely lead to growing conflicts over mining resources and rights (Krause et al. 2022; Veiga and Marshall 2019). Although conflicts can drive democratic processes (Rowbotham 1998), achieving equity in mining territories is necessary to prevent violence-related conflicts and move toward more sustainable pathways. Furthermore, because power dynamics influence interpretations of sustainability and equity (Boonstra 2016), understanding power's effect on sustainability transitions is essential for addressing why inequity in gold mining territories prevails (Bedford et al. 2020).

Our research in the Colombian Amazon examined issues of equity and sustainability in gold mining areas. The Colombian Amazon is known for its biodiversity and complex socio-political landscape. After the 2016 peace accord, a power vacuum led to increased legal and illegal gold mining. This activity involves overlapping claims among states, companies, artisanal miners, environmental groups, and post-conflict factors. Our findings highlight how local conflicts over titling, legality, and recognition in mining reflect broader tensions related to three different narratives to transitioning to more sustainable pathways: Greening gold mining, Assisting artisanal and small-scale mining (ASGM), and Emancipatory practices of resistance. The "Greening mining" narrative emphasizes advanced technology and corporate responsibility, but in practice, it does little to address the unequal relationship between mining industries and local communities. The ASGM assistance narrative demands transparency and recognition of traditional rights, while Indigenous groups and other local actors advocate for alternatives to Western models of modernization and sustainability.

Corporate responsibility is the set of practices through which mining corporations aim to meet stakeholders' expectations beyond shareholders (Jamali and Mirshak 2007; Karakaya and Nuur 2018). However, corporate responsibility narratives often reinforce the legitimacy of modern corporations as decision-makers capable of (re)distributing mining's benefits and burdens (Dorn 2022; Jamali and Mirshak 2007). This has effectively displaced debates around mining's contributions to the Amazon's social-ecological degradation and the climate crisis (Bedford et al. 2020; Dorn 2022). Despite evidence that mining royalties contribute minimally to public revenues, and state support often subsidizes private investment through its tax system (Vélez-Torres 2014), corporate actors maintain dominant positions in sustainability discussions in the country. These asymmetries raise questions about the role of corporate mining in sustainability transitions, particularly given the projected rise in global gold demand (Trench et al. 2023).

In our case study, tackling these widespread power imbalances calls for action on two fronts. First, integrating ASGM into sustainability discussions at national and regional levels is crucial, recognizing miners' rights, supporting autonomous livelihoods, and fostering negotiated coexistence rather than assimilation into corporate supply chains (Diaz et al. 2020; Güiza-Suárez and Kaufmann 2024; Hilson et al. 2020; Pedersen et al. 2021; Sauerwein 2023). Second, scholars, practitioners, and policymakers must confront the (3D) epistemic domination of Western knowledge systems over Indigenous and local epistemologies to avoid reinforcing epistemic injustices (Bryant 2015; Escobar 2018).

In practice, more pluralistic governance models are needed to create sustainable mining futures in the Colombian Amazon (Caniglia et al. 2020; Ministerio de Minas y Energía and Agencia Nacional de Minería 2023; World Bank 2024). Participatory monitoring mechanisms, autonomous co-management arrangements, and plural legal frameworks that respect statutory and customary rights could help redistribute power and authority more equitably (Caniglia et al. 2020; Tengö et al. 2014). Examples of the latter are the Bolivian and Ecuadorian constitutions, which recognize their countries as plurinational states, allowing for more inclusive legal recognition of Indigenous governance

systems and ancestral land rights within national legal frameworks. These approaches should not merely document local values but actively create spaces for cross-pollination, learning, and negotiation among marginalized communities, researchers, and policymakers (Zafra-Calvo et al. 2020).

Furthermore, despite frequent conflicts between ASGM and corporate actors over territory, legality, and access to resources, Colombia has few formal mechanisms to resolve these disputes locally. Literature emphasizes the value of multi-stakeholder platforms, negotiated access frameworks, and rights-based formalization paths as tools to decrease conflict and promote fair coexistence (Hilson et al. 2021; Verbrugge and Besmanos 2016; World Bank 2024). Yet, our research indicates that these mechanisms are largely absent, where ASGM actors are often excluded from formal consultations and legal reforms, resulting in increased distrust and opposition.

Decoloniality plays a critical role in reimagining sustainability transitions in the Colombian Amazon. As seen in emancipatory practices and resistance, decoloniality emphasizes the creation of alternatives to development rather than alternatives for development (Escobar 2018). It involves using local languages, valuing Indigenous knowledge systems, and promoting concepts like Buen Vivir as legitimate foundations for decision-making in the region. For instance, several interviewees explicitly challenged state-led formalization efforts as misaligned with their lived realities and values, viewing them as forms of epistemic erasure rather than inclusion. Decolonizing sustainability practices requires guarantees for the autonomy of local communities, recognition of both customary and statutory mining rights, and control over territories that have been historically marginalized by the state and exploited by corporate interests. Such calls for territorial self-determination and knowledge sovereignty emerged from both Indigenous leaders and ASGM actors, who resist being treated merely as objects of intervention. Recognizing Indigenous and ASGM actors as political and epistemic agents, not just beneficiaries, is crucial for fairer, inclusive sustainability in the Colombian Amazon. These perspectives enhance justice-focused approaches by highlighting both socio-technical options and epistemic and ontological diversity rooted in territorial struggles.

While some aspects of our case, such as Colombia's history of armed conflict, are specific to that context, the broader patterns of knowledge exclusion and disputed legitimacy are relevant to extractive settings worldwide. Therefore, our findings contribute to debates on sustainability transitions by demonstrating how structural exclusions, epistemic hierarchies, and relational power dynamics shape the possibilities for equitable change in socio-ecological systems. As emphasized in recent transitions research, sustainability transformations require more than technological innovation or institutional reform—they demand shifts in power relations and the inclusion of historically marginalized actors and knowledges (Avelino 2021; Caniglia et al. 2020; Scoones et al. 2020).

In gold mining governance in the Colombian Amazon, dominant narratives centered on technological greening and corporate responsibility often reproduce existing hierarchies, limiting the transformative potential of sustainability efforts. In contrast, emancipatory practices and Indigenous resistance exemplify forms of bottom–up transition that challenge Western-centric development paradigms and expand the scope of what constitutes sustainable development**.** This aligns with calls in the sustainability transitions literature to pay closer attention to informal, peripheral, and contested spaces of change (Coenen et al. 2012; Stirling 2011), and to foreground justice, recognition, and plurality in imagining sustainable futures (Feola 2020; Köhler et al. 2019; Scoones et al. 2020). From this perspective, transformation involves not only socio-technical changes but also fundamental shifts in whose knowledge is valued, whose voices are listened to, and who controls resources and territories.

Conclusions

Precious metals, such as gold, are critical for developing new technology that can utilize renewable forms of energy. Finding less harmful ways to mine these metals is a key focus of discussions and solutions toward sustainability transformations. To contribute to this effort, we conducted a discourse analysis on narratives of sustainable gold mining in the Colombian Amazon. Our findings reveal that various actors involved in Colombian gold mining interpret the concept of sustainability in different ways. While the greening of mining involves a corporate focus on distributive equity, traditional and artisanal miners seek support and recognition of their mining rights. Indigenous, ethnic minorities, and environmental organizations aim to emancipate themselves from historical (3D) epistemic domination and (2D) exclusion in mining territories. Competing interests and values in mining territories have led to (1D structured and 2D structural) conflicts. Our research suggests addressing these conflicts by undoing unequal exchanges between corporate actors, state agencies, miners, and Indigenous and local people. This involves questioning the role of corporate mining in sustainability transitions and diversifying ideas of good living. Epistemic equity and the construction of power that is perceived as legitimate are critical initial steps to transition to more sustainable pathways in the Colombian Amazon.

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Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

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